

Detecting Outliers in Time Series Using Window-Sliding *ARIMA*: A Case Study on Montreal’s Daily Temperatures

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Abstract

This article describes and implements a method for outlier detection in the specific context of time series, drawing on the taxonomy proposed by Blazquez-Garcia *et al.* (2021). First, the study briefly presents the aforementioned taxonomy, then focuses on anomaly detection in daily temperature data from Montreal, Canada. The aim is both to implement a modern outlier detection technique and to carry out this analysis on data with high-stakes implications: those related to climate change. For this, we use a historical dataset on Canadian climate. This database contains daily information on average temperatures and precipitation for 13 Canadian weather stations from 1940 to 2019. The detection method is carried out using an *AutoRegressive Integrated Moving Average (ARIMA)* model with a sliding window. It detects an increasing number of outliers over time, which suggests, to some extent, the existence of climate warming. This article is the first to combine a little-used method with climate data from Montreal.

1 Introduction

Main definitions

Let $T \subset \mathbb{Z}$ be a set of time indices, and let $(X_t)_{t \in T}$ be a stochastic process defined on a probability space Ω with values in $\mathcal{X}$. A *time series* is a realization of this process, that is, for $\omega \in \Omega$, a mapping $t \mapsto X_t(\omega)$ for all $t \in T$, also denoted as $\{x_t\}_{t \in T}$.

A class of well-known models for stochastic processes is that of *autoregressive models*, including the *ARIMA* family, defined as follows.

Let $(X_t)_{t \in T}$ be a stochastic process taking values in $\mathbb{R}$. Let $p, d, q \in \mathbb{N}$, $c \in \mathbb{R}$, and $\alpha_1, \dots, \alpha_p \in \mathbb{R}$, $\beta_0, \dots, \beta_q \in \mathbb{R}$. Let $(\varepsilon_t)_{t \in T}$ be a strong white noise process, *i.e.*, a sequence of independent and identically distributed random variables with zero mean and finite variance.

The process $(X_t)_{t \in T}$ is said to follow an $ARMA(p, q)$ model if:

$$X_t = c + \sum_{i=1}^p \alpha_i X_{t-i} + \sum_{j=0}^q \beta_j \varepsilon_{t-j}$$

It follows an $ARIMA(p, d, q)$ model if the process has been differenced d times to achieve stationarity before fitting an $ARMA(p, q)$ model.

Related work and contribution

The analysis of time series can serve in particular two objectives: the detection of outliers using new methods, and the application to meteorological data that may or may not confirm the presence of climate warming.

Some authors already demonstrate the presence of climate warming using *ARIMA* or *Seasonal AutoRegressive Integrated Moving Average (SARIMA)* models, such as Kajuru (2019) or Asamoah-Boaheng (2014), who respectively analyze temperature in Zaria, Nigeria, and in the Ashanti region in Ghana. Specifically, Kajuru concludes with a $SARIMA(1, 0, 1)(0, 1, 1)_{12}$, while Asamoah-Boaheng obtains a $SARIMA(2, 1, 1)(1, 1, 2)_{12}$. The use of *SARIMA* is also complemented by preliminary tests for detecting long-term trends, using the Mann-Kendall test and Sen’s slope (Kabbilawsh *et al.*, 2020). Similarly, the analysis conducted by Sadhukhan *et al.* (2022) applies these same tests on 10-year periods and reveals climate warming starting in 1950 of 0.0002°C per month.

Moreover, some authors complicate the training of *ARIMA* models by using sliding windows, that is, segmenting the data chronologically and shifting the windows by a certain period at each iteration. Ma *et al.* (2017) and also Zhou *et al.* (2018) apply this method respectively to meteorology and oceanography to detect outliers, and obtain good results; whereas Sheoran & Pasari (2022) apply it to wind speed data from 4 Indian cities. They show that *Window-Sliding ARIMA (WS-ARIMA)* reduces the root mean square error by up to 75% for daily wind speed data and 50% for weekly data.

Finally, non-linear methods are also used (Viola *et al.*, 2010) to evaluate the dynamics of climate warming across several cities, in particular in Montreal. Their analysis also accounts for the inevitable noise present in the data, using robust methods. Temperatures are predicted up to the year 2028, and tests are conducted on known periods to validate the models.

This state of the art reveals that outlier detection in time series mainly employs methods based on autoregressive models (*ARIMA*, *SARIMA*, *WS-ARIMA*) and are applied to meteorological data. However, no article combines outlier detection within Canadian data and the use of *ARIMA* models with sliding windows. This article therefore contributes by exploiting climate data from Montreal, Canada, in order to detect anomalies in temperatures, which may indicate the presence of climate warming.

In a first section, we summarize the article on which our approach is based, in order to present the different types of anomalies we might encounter and the various methods used to detect them

in time series. We will then detail a specific approach used for anomaly detection in time series — an *ARIMA* model with sliding windows — before applying it to our Canadian data.

2 Summary of the article

For our analysis, we relied mainly on the article *A Review on Outlier/Anomaly Detection in Time Series Data* by BLAZQUEZ-GARCIA *et al.* (2021) as a clear baseline and framework. This article provides a thorough literature review on methods for detecting outliers specifically in time series data. Its goal is to build a taxonomy that distinguishes between types of outliers, the input data types, and the type (or subtype) of detection methods used.

It's important to note that the precise definition of outliers is not universally agreed upon, although Hawkins' definition: "An observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism" (1980 [1]) is often used as a starting point. The actual definition varies depending on the type of series considered.

Here, focusing on time series data, the review looks at identifying *outliers* regardless of the terminology authors might use (*anomalies*, *discordant observations*, *discords*, *exceptions*, *aberrations*, *surprises*, *peculiarities*, *contaminants*), and within an unsupervised setting, i.e., considering each variable equally without linking one or more variables to a label variable to predict or explain. Moreover, the series are assumed to be regularly sampled over time (the gap between two consecutive points is constant throughout the series).

The input data can be univariate or multivariate time series.

The article then considers three types of outliers:

- *Type I Outliers* or *point anomalies*. These are outliers that behave unusually at a specific point in time, compared either to their neighborhood values (called *local* outliers) or the entire series (called *global* outliers).
- *Type II Outliers* or *subsequence anomalies*. These are outliers whose joint behavior is unusual, even if individual observations are not anomalous. These can also be local or global.
- *Type III Outliers*¹ or *anomalous time series*. This applies only when the input data are multivariate and refers to entire series whose behavior differs significantly from other series considered.

Mathematically, an outlier can be described as:

- A single point or vector in the series (Type I),
- A subset of points or vectors in the series (Type II), where the size, called *length*, is usually set by the analyst,

¹Although this naming is not explicitly given

- A univariate series (Type III).

This gives us five possible analysis scenarios:

1. Univariate data – Type I outliers (*)
2. Multivariate data – Type I outliers
3. Univariate data – Type II outliers (**)
4. Multivariate data – Type II outliers
5. Multivariate data – Type III outliers

Once the data type and outlier type are determined, the article proposes a set of methods for each case:

1. **Univariate data – Type I outliers**

- ▷ Model-based methods: divided into estimation and prediction methods,
- ▷ Density-based methods,
- ▷ Histogramming methods.

2. **Multivariate data – Type I outliers**

- ▷ Model-based methods: divided into estimation and prediction methods,
- ▷ Dissimilarity-based methods,
- ▷ Histogramming methods.

3. **Univariate data – Type II outliers**

- ▷ Discord detection methods,
- ▷ Dissimilarity-based methods,
- ▷ Prediction model-based methods,
- ▷ Frequency-based methods,
- ▷ Information theory-based methods.

4. **Multivariate data – Type II outliers**

- ▷ Model-based methods: estimation and prediction,
- ▷ Dissimilarity-based methods.

5. **Multivariate data – Type III outliers**

- ▷ Dimensionality reduction methods,
- ▷ Dissimilarity-based methods.

Other characteristics mentioned in the article help describe outlier detection methods:

- The goal of the analysis: outliers can be seen either as unwanted data (to be removed to clean the dataset) or as interesting events (to be detected and analyzed).
- Temporality: whether the method uses the temporal order of observations or not. This often affects the method’s validity or performance if the series is permuted.
- Streaming capability: whether the method can handle data in real-time, processing new points as they arrive.
- Parameterization: whether the method is parametric, semi-parametric, or non-parametric.
- Dimensionality: univariate or multivariate approach. In multivariate input data, sometimes the problem is simplified to a univariate one.

Overall, the methods reviewed in the article share these features:

- Concern regularly sampled time series data,
- Are unsupervised,
- Consider outliers as either contaminants or events of interest,
- Use univariate or multivariate data,
- Detect Type I, II, or III outliers,
- Use temporal information or not,
- Employ sliding windows or not (local/global analysis),
- Support streaming or not,
- Are parametric, semi-parametric, or non-parametric,
- Use univariate or multivariate techniques.

3 Data

3.1 Database Overview

In order to carry out this project, we selected a Government of Canada database entitled ”*Eighty years of Canadian climate data*”. This database compiles daily average temperature and precipitation data from 13 Canadian weather stations, covering the period from 1940 to 2020. For some stations, data are only available starting in 1960.

For this study, we focus on data from Montréal over the period from January 1, 2000, to December 31, 2019. This dataset contains 7,244 observations, of which 0.85% (62 observations) are missing. These missing values are distributed throughout the period and must be addressed in our analysis.

3.2 Handling of Missing Values

Temperatures exhibit significant seasonal variations, which makes inappropriate replacing missing values with a fixed value such as the median. We therefore chose to impute missing values using the temperature recorded on the previous day.

In the rare cases where no prior temperature was available, two scenarios were identified:

- For 2014 and 2018, the missing values were manually replaced based on trends observed in neighboring data.
- In all other cases, the observations were removed to avoid bias.

This approach preserves the temporal coherence of the dataset as much as possible while avoiding distortions in its distribution due to imputation.

3.3 Descriptive Statistics

The focus of this study is the daily average temperatures in Montréal, which range from -26.6°C to 30°C , with a mean of 7.41°C (Table 1). Negative temperatures are strongly represented, as indicated by the first quartile at -1.7°C . Conversely, extremely high average temperatures are less frequent, with a maximum of 30°C and a third quartile of 17.8°C .

Table 1: Descriptive Statistics

Mean	1st Quartile	Median	3rd Quartile	Standard Deviation	Min	Max
7.41	-1.7	8.2	17.8	11.83	-26.6	30

The entire dataset can then be represented graphically (Figure 1). A clear seasonality and a recurring yearly trend can be observed, which is expected for temperature data. This seasonality must be taken into account when detecting anomalies. Minimum and maximum temperatures fluctuate slightly across years but do not appear to follow a defined trend.

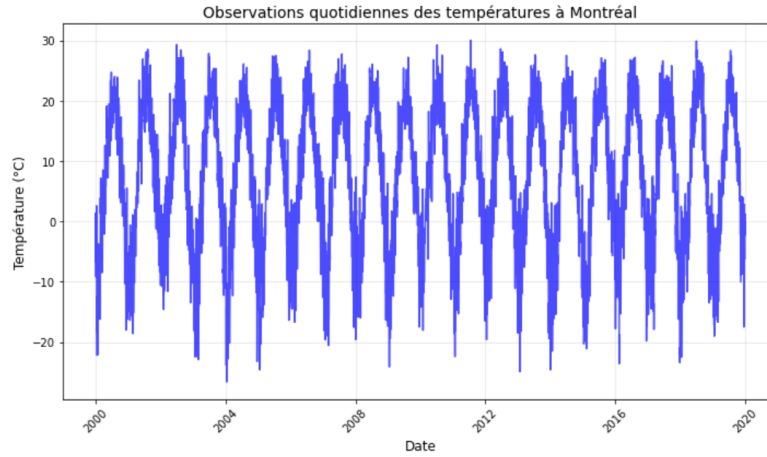


Figure 1: Daily average temperatures in Montréal between 2000 and 2020

We then compute the annual mean temperatures (Figure 2), which can also be represented graphically. The mean fluctuates regularly between 6.5°C and 8.5°C , reflecting a relatively high level of variability in this context. Once again, no clear trend is observed.

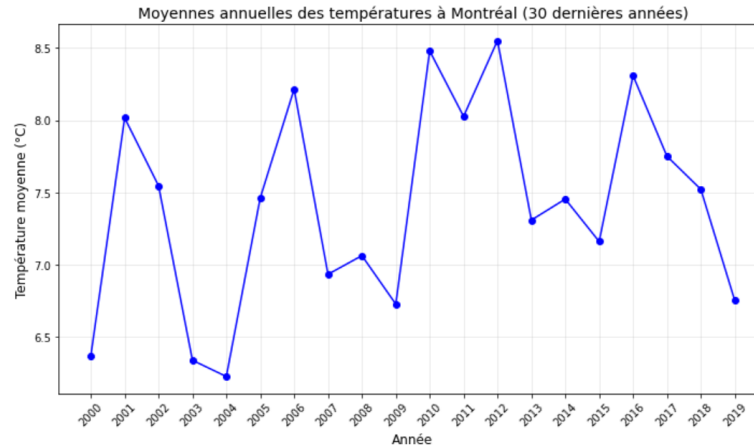


Figure 2: Annual mean temperatures in Montréal between 2000 and 2020

Finally, we plot a scatterplot of the daily average temperatures (Figure 3). A strong concentration of points is observed around 10°C , while the data become more dispersed at the extremes, particularly at lower temperatures. These distant observations could potentially be considered as outliers. However, factors such as seasonality must be taken into account, and the analysis will vary depending on the type of anomalies under investigation (single points versus sequences).

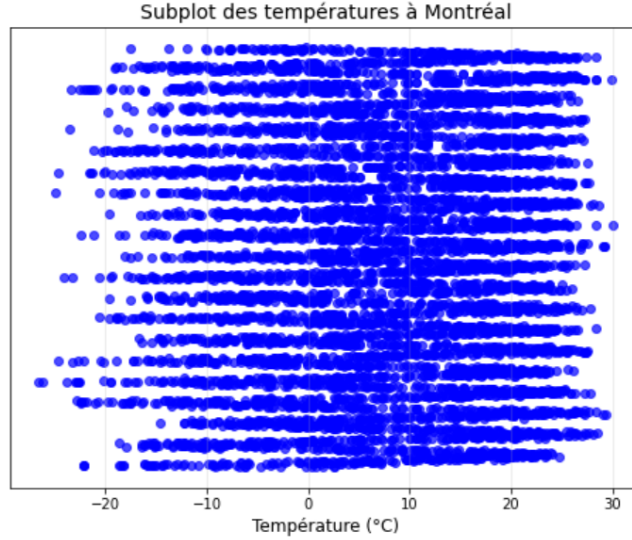


Figure 3: Scatterplot of daily average temperatures in Montréal between 2000 and 2020

4 ARIMA Method with Sliding Window

In this section, we present a method that falls within the first analytical framework [??] (marked with (*)), which aimed at detecting Type I outliers using univariate data. As discussed earlier, this first framework includes several methods, among which we selected a prediction-based approach. Specifically, we implement a sliding-window *ARIMA* model manually² in `Python`, based on the method proposed by Zhou et al. (2018), described in pseudo-code. In their article, Zhou et al. apply an *ARIMA* model to seafloor data without accounting for potential seasonality. Here, since seasonality is clearly present in our meteorological data, we illustrate the method on a single year, namely 2019.

4.1 Methodology

The general approach is as follows: a window is selected, the best *ARIMA* model is determined, the values of the series and its future values are predicted, and each observation is then labeled as “outlier” or not based on the difference between the predictions and the observed data. This process is iterated by “sliding” the window across the series. Each iteration may thus reveal outliers that could have been overlooked if the analysis were performed in a single step.

More precisely, the procedure is as follows:

- Select a window, i.e. a time-ordered subsample of the series of length *wind_length*
- Identify the *best ARIMA(p, d, q)* model by choosing parameters *p*, *d*, and *q* according to the AIC criterion

²No pre-built package is available

- Estimate the selected model
- Predict the values, denoted $\hat{x}_t$, for the entire series:
 - For values within the window: perform *in-sample* predictions
 - For subsequent values (until the end of the year): perform *out-of-sample* predictions
 - For earlier values (from the start of the year to the beginning of the window): leave these as missing, which does not hinder the analysis
- Compute the distance between predictions and observed values x_t and determine whether this distance exceeds a predefined threshold τ , i.e., if

$$d(x_t, \hat{x}_t) > \tau$$

where d denotes the Euclidean distance.

- Label each observation as "True" if identified as an outlier, and "False" otherwise

The process is then repeated by sliding the window with a step size *shift_step*, i.e., selecting a new subsample of the same length, starting a fixed number of days later.

4.2 Illustration

We illustrate this method using daily average temperatures in Montréal in 2019 (365 observations), with the following parameters: *wind_length* = 60, *shift_step* = 40, and $\tau = 8$ (Figure 4). It is clear that the sliding-window approach (red points) identifies more outliers than a global method (purple crosses). However, detection is highly sensitive to the parameter choices, particularly the threshold used to define abnormal observations. Expert judgment would therefore be more appropriate.

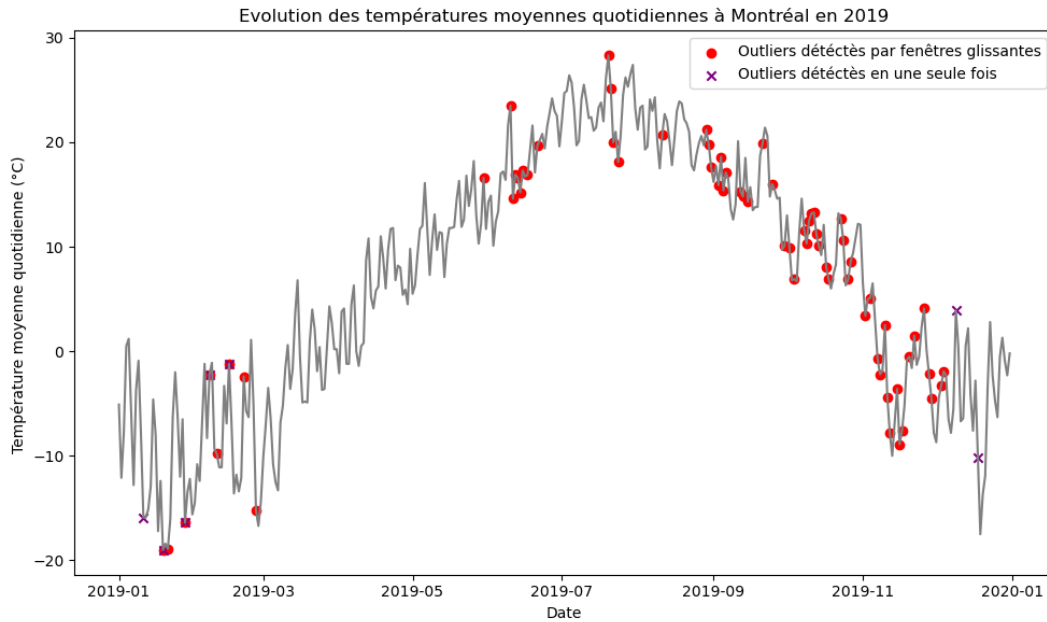


Figure 4: Outlier detection using sliding windows

Conclusion

The goal of this project was to detect anomalies in a temperature dataset from the Canadian government using an innovative method. This is an important step in meteorological data analysis because it can help identify abnormal average temperatures, which may be an indicator of climate change. The aim is therefore more to provide an overview of abnormal temperatures rather than to detect possible errors in the database. To do this, we used a prediction-based method: the ARIMA method with a sliding window, which allows detecting Type I outliers, *i.e.*, point anomalies. This method tends to detect many outliers, especially because the first windows do not predict the values well toward the end of the series. However, it makes it possible to detect outliers that might have been hidden when performing a global analysis, *i.e.*, predicting the entire series all at once. The analysis could be complemented by using models that take seasonality into account (SARIMA models), which would allow more relevant predictions and consequently, the inclusion of data from other years.

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